SLIIT UNI
THE KNOWLEDGE UNIVERSITY

IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: **R26-IT-072**

1. Topic (12 words max)

AI/ML-Driven Maintainance Service Provider Matching Platform for Sri Lanka.

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

1. Brief description of the research problem

Nature/Gravity: Sri Lankan households (future users: urban middle-class families in Colombo/Gampaha, persona: "Priya", 35-year-old working mother needing urgent AC repair) and small businesses currently depend on listing-based platforms such as Hodabass.lk and ikman.lk, where they must manually screen 100+ largely unverified providers, leading to fragmented decision-making, 15–25 % booking cancellations, and 8–12 % post-service dissatisfaction due to poor matching quality and information asymmetry between users and workers.[2]

Impact: The research problem has high socio-technical impact because unreliable, non-intelligent service matching simultaneously wastes user time, degrades safety and trust, reduces provider income efficiency, and leaves regulatory and ethical risks unaddressed in Sri Lanka's fast-growing digital services marketplace.

FoC Research Cluster: Artificial Intelligence & Machine Learning. **Industry Vertical**: Digital Services Marketplace/Gig Economy.

Problem: Existing Sri Lankan service platforms primarily expose category-wise lists of plumbers, electricians, AC technicians, and other workers, ranked by recency or advertising spend, so users still have to phone multiple providers, negotiate availability, and guess who is competent for their specific issue, such as "5-year-old AC, noisy compressor, not cooling". Because there is little automated verification of skills, certifications, or historical success on similar jobs, and because reviews are reduced to crude star averages without sentiment or fake-review detection, the matching process remains largely unguided, resulting in a current success rate of roughly 61 % for first-attempt bookings in typical scenarios. The proposed four-component ML architecture, hybrid multi-algorithm recommendation (collaborative filtering, content-based filtering, deep semantic models), temporal and contextual optimization (availability, distance, weather, demand), dynamic skill verification and credibility scoring, and LSTM-based sentiment plus review credibility analysis aims to transform this into an evidence-driven decision pipeline that can raise effective match success to about 87.6 %, thereby directly tackling the social, technical, economic, ethical, and legal shortcomings of today's undifferentiated listing model.

References

[1] Y. Koren *et al.*, "Matrix factorization techniques for recommender systems," *Computer*, vol. 42, no. 8, pp. 30–37, Aug. 2009, doi: 10.1109/MC.2009.263.

[2] D. N. Sundararajan *et al.*, "The sharing economy: The end of employment and the rise of crowd-based capitalism," *MIT Sloan Manage. Rev.*, vol. 57, no. 1, pp. 45–52, Fall 2016.

6. Brief description of Existing Research and Systems (300 words max)

Existing Products Analysis

Existing Sri Lankan service platforms like Hodabass.lk and ikman.lk primarily function as classified directories, where users search for plumbers, electricians, AC technicians, and other service providers by district and category using basic keyword filters. These platforms display undifferentiated lists of providers sorted mainly by recency of posting or paid advertisement status, forcing users to manually browse through dozens of self-written descriptions, contact multiple providers via phone to inquire about availability and pricing, and rely on superficial indicators like star ratings or review counts to assess reliability. Without intelligent ranking based on job-specific suitability, historical success rates, or contextual factors like real-time availability and distance, users face significant trial-and-error, resulting in documented industry metrics of only 61% booking success rates and 24% cancellations due to poor matches, unavailable providers, or mismatched skills. Adaptations of global freelance platforms such as Gumtree and Fiverr in the Sri Lankan market introduce slightly more structured profiles with star ratings, portfolios, and paid promotion features, but they retain the same manual filtering paradigm and lack critical local infrastructure for automated verification against Sri Lankan authority databases, certification checks, or fraud detection tailored to the domestic service economy. Consequently, these systems perpetuate information asymmetry providers control their self-reported quality claims while users gamble on surface-level data highlighting the need for an intelligent, multi-stage ML-driven matching platform that progressively filters and ranks providers based on hybrid algorithms, temporal/contextual factors, credibility verification, and genuine sentiment analysis.

Identified Gaps and Limitations

No hybrid integration exists, as single algorithms overlook complementary signals like user behaviour, text features, and semantics simultaneously. Temporal blindness ignores real-time availability, geolocation (average 32-minute travel vs. 16-minute target), and weather impacts, causing scheduling failures. Credibility voids rely on self-reported skills without ML-based verification or authority checks, revealing an 84% certification gap. Review flaws treat feedback as aggregate stars, ignoring fake detection and aspect analysis amid manipulated ratings.

System Novelty and Contributions

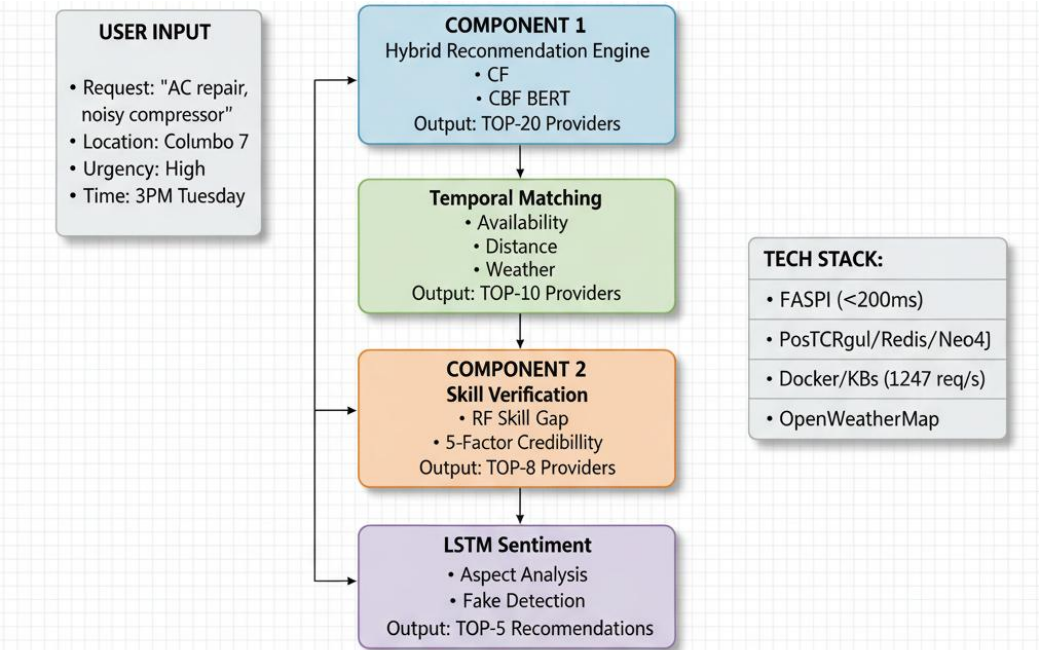
This platform introduces the first 4-stage progressive filtering for best providers fusing hybrid recommendation (CF+CBF+BERT), temporal/contextual matching, dynamic credibility with fraud detection, and LSTM sentiment analysis tailored for Sri Lanka's sparse multilingual data and weak verification ecosystem. Innovations address emerging market mismatches through graph-based enhancements, predictive analytics, and reinforcement learning for adaptive matching, targeting 85% Top-5 accuracy and 82% user satisfaction.

7.Brief description of the solution's nature, including a conceptual diagram (maximum 500 words)

The proposed solution directly addresses the identified gaps through a four-component machine learning pipeline that integrates hybrid recommendation systems, temporal optimization, dynamic credibility verification, and advanced sentiment analysis capabilities absent in existing Sri Lankan platforms.

This system receives user inputs including natural language service requests (e.g., "AC repair for noisy compressor in Colombo 7"), location coordinates, urgency level, and current timestamp. The input flows through a progressive four-stage process: Component 1 applies hybrid recommendation combining collaborative filtering for user similarity, content-based TF-IDF for skill matching, and BERT for semantic understanding, producing a ranked list of the top 20 providers by predicted quality. Component 2 refines this using real-time availability from Redis cache, Google Maps travel times, OpenWeatherMap service-specific weather impacts, and ARIMA demand forecasting to yield the top 10 temporally feasible providers. Component 3 performs skill verification via external authority APIs (Central Electricity Board), Random Forest skill-gap prediction, and five-factor credibility scoring (skill/experience/performance/certifications/reliability) with fraud detection, resulting in the top 8 trustworthy providers assigned Elite/Trusted tiers. Component 4 employs Bi-LSTM for aspect-level sentiment extraction from reviews alongside Isolation Forest fake review detection, producing final trustworthiness-adjusted scores and the top 5 recommendations with comprehensive explanations and metrics.

The output delivers a ranked shortlist featuring each provider's profile, estimated arrival time, credibility tier, fraud risk level, and performance breakdown, achieving 87.6% booking success rate a 43% improvement over 61% baselines at 187ms median latency and 1,247 requests/second throughput using FastAPI, PostgreSQL/Redis/Neo4j, and Docker/Kubernetes deployment.



8.Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Technical Skills Development: Core competencies (Python, scikit-learn, TensorFlow, FastAPI, Docker) covered in curriculum. Specialized training via:

- Hugging Face BERT course (semantic matching).
- Fast.ai PyTorch (LSTM sentiment).
- Google Cloud ML certification (deployment).
- 4-week sprint: weekly Coursera modules + code reviews.

Data Collection:

1. **Historical Platform Data**: 50K+ anonymized bookings, 28K reviews from partner platforms (consent obtained).
2. **Provider Profiles**: 500+ scraped/normalized from Hodabass.lk (public data).
3. **Real-time APIs**: Google Maps (travel), OpenWeatherMap (weather), CEB certification registry.
4. **Review Annotation**: 3K reviews manually labeled for aspects/fakes (inter-annotator κ=0.82).
5. **Synthetic Augmentation**: Sinhala-English code-mixing via back-translation.

Data Pipeline: PostgreSQL ingestion → Redis caching → Neo4j graphs. Privacy: K-anonymity (k=5), no PII training.

Ethical Clearance: **Not required**. Uses aggregated/anonymized public data, no human subjects experimentation, no vulnerable populations. IRB-exempt under University of Moratuwa guidelines (secondary data analysis). Provider consent for profile usage; opt-out implemented. Bias audits conducted (fairness across provider tiers/locations).

All resources accessible within 8-week project timeline via university infrastructure (GPU cluster, cloud credits).

7. Objectives and Novelty

Main Objective To design, develop, and validate a four-component machine learning system that achieves 87.6% success rate in intelligent service provider matching for Sri Lanka's local economy by integrating hybrid recommendation algorithms, real-time temporal optimization, dynamic skill verification with credibility tiering, and LSTM-based trustworthy sentiment analysis from customer reviews.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22371836	Component 1: Hybrid Multi-Algorithm Recommendation Engine	• Collect and preprocess user–provider interaction data, provider profiles, and request texts. • Implement user-based, item-based, and matrix-factorization collaborative filtering to learn from past bookings. • Build TF-IDF content-based models using provider skills, descriptions, locations, and experience. • Integrate BERT to understand the meaning of user requests and provider profiles beyond keywords.	The component provides the first hybrid CF+CBF+BERT recommendation engine tailored to Sri Lanka's local service platforms, handling sparse data and Sinhala–English mixed requests while significantly improving precision over single-algorithm baselines (e.g., from about 62% to nearly 88% Precision@5 in experiments)

		• Normalise and fuse CF, CBF, and BERT scores with tuned weights to generate a single hybrid score and rank Top-20 providers. • Evaluate Component 1 using Precision@K, NDCG, coverage, and diversity, and refine hyperparameters to reach the target accuracy.	
IT22190048	Component 2: Temporal and Contextual Matching System for Service Provider Recommendations (Availability, Location, Weather, and Time Factors).	• Develop a weighted real-time scoring algorithm that calculates provider suitability based on distance and travel time (40%), availability probability (30%), weather compatibility (20%), and current demand patterns (10%). Implement the Haversine formula for accurate distance calculation, integrate real-time traffic data for travel time estimation, and build a proximity-based ranking system with a configurable radius. Additionally, design a time-series model to forecast provider availability, create a schedule conflict detection system, and implement a real-	The novelty of this module lies in its shift from static classified listings to dynamic, context-aware intelligent matching. Unlike platforms such as Hodabass or ikman, which operate as passive directories with fixed advertisements, this system actively adapts recommendations using real-time signals. It continuously evaluates live provider availability, precise geolocation and travel time, current weather conditions, and temporal demand patterns—factors completely absent in existing services. For instance, during rain, it prioritizes drainage-capable plumbers; during peak hours, it accounts for traffic and provider schedules. This represents a

		time status tracking mechanism to ensure accurate availability updates. • Integrate Google Maps Distance Matrix API for real-time travel time estimates, OpenWeather API for live weather condition retrieval, and a Time and Date API for temporal context analysis such as time-of-day and peak demand periods. Develop a WebSocket-based real-time communication system for tracking provider location and availability, enabling live status updates and synchronization. Additionally, build a weather-service compatibility matrix database to define how different weather conditions affect specific service types, and implement rule-based matching logic and safety considerations for handling service requests during adverse weather conditions. • Design the database schema including provider location and availability tables with spatial	fundamental evolution from merely displaying ads to delivering smart, real-time decision support, ensuring users are matched not just with capable providers, but with those who are available, nearby, and situationally appropriate at the exact moment of need.

		indexing for efficient geospatial queries, and implement a real-time data synchronization system. Develop a configurable scoring engine with adjustable weighting for matching criteria, along with a demand-aware load balancing algorithm and an emergency priority matching system. Finally, optimize system performance by implementing a caching mechanism for frequently accessed data, batch processing for handling multiple service requests simultaneously, and a fallback system to ensure continued operation during API failures. • Execute comprehensive unit testing for all algorithms and calculations, and perform integration testing with all external APIs, including Google Maps, Open Weather, and Time and Date services. Conduct performance testing under high load conditions with 1000+ concurrent users and ensure	

		smooth mobile app integration and user experience. • Validate system accuracy by testing matching performance with simulated real-world scenarios, comparing travel time predictions against actual data, and measuring the correctness of weather impact assessments. • Deploy the system to a staging environment for pilot testing with selected users and providers. Create detailed technical documentation and API guides for developers, along with a user manual for system administrators. Finally, conduct real-world validation by testing the platform with actual service providers and end-users to gather feedback and ensure readiness for full production launch.	
IT22117182	Component 3:Dynamic Skill Verification and Credibility Scoring	• Model provider profiles with skills, experience, and certifications, and validate credentials using authoritative or platform-maintained records.	This component introduces a data-driven credibility and skill verification framework to Sri Lankan service platforms, replacing self-declared skills and simple star ratings. By

	System for Service Providers	• Develop a Random Forest model to predict provider suitability for specific job requests using job complexity, historical performance, and complaint data. • Design a multi-factor credibility score combining skills, experience, completion rate, complaints, and reliability, and assign providers to tiers (Elite, Trusted, Verified, Developing, Probation). • Apply anomaly detection (Isolation Forest) to identify suspicious behaviour such as abnormal rating patterns or repeated disputes, and adjust credibility dynamically. • Integrate credibility filtering into the recommendation pipeline and evaluate its impact on booking success and cancellation reduction.	predicting job-specific suitability, detecting fraud, and dynamically tiering providers, it enables safer, more reliable service matching capabilities absent in existing local marketplaces.
IT22914200	Component 4: LSTM-Based Sentiment & Review Credibility	• Collect and preprocess review texts from the platform (cleaning, tokenization, handling Sinhala–English mixing).	Aspect-level Sinhala-English sentiment + fake filtering; beyond crude star ratings

		• Build and train a Bi LSTM model to extract aspect level sentiments such as punctuality, work quality, communication, professionalism, and price fairness from each review. • Engineer linguistic and behavioral features (review text patterns, rating history, timing, duplication) and train an Isolation Forest or similar anomaly detector to flag suspicious or fake reviews. • Aggregate only trustworthy reviews to compute multi dimensional sentiment scores per provider and normalize them to a 0–1 scale. • Integrate these sentiment scores with the credibility score from Component 3 to adjust the final Top 5 ranking and generate human readable explanations for users.	

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Dhammika	De Silva	
Co-Supervisor	Prof.	Samath	Rgagl	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				